



**Drawing and testing assumptions**

# **Parametric and non-parametric hypothesis tests**

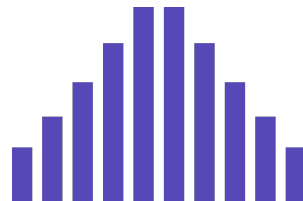
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# Statistical tests overview

In hypothesis testing, **statistical tests** are used to decide whether we **reject or fail to reject the null hypothesis**. We use either **parametric** or **non-parametric tests**, depending on some factors of the underlying data.

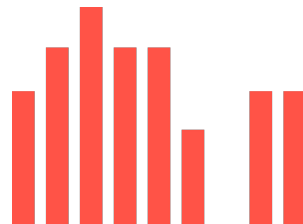
## Parametric tests

- **Assume** that the data follow a **specific distribution**, such as a normal distribution.
- **More precise** estimates of the population parameters **when the assumptions are met**.
- When the data **violate the assumptions**, these tests may produce **inaccurate or unreliable** results.



## Non-parametric tests

- **No assumptions** about the distribution of the data.
- **More robust to violations** of the assumptions but less powerful when the assumptions of the parametric test are met.
- Often preferred when the **sample size is small**.



# Parametric tests overview

**Parametric tests** are a group of statistical tests that are used to **test hypotheses** on population parameters, such as the mean or variance, by making certain **assumptions about the underlying distribution of the data**.

## T-test

Used to test hypotheses on the mean of a **single** population or the **difference** between the **means** of two populations with **small sample sizes**.

## Z-test

Similar to the t-test, the z-test assumes that the population **standard deviation is known** and the **sample is large**.

## F-test

Used to test hypotheses on the **difference** between the **variances** of two or more populations with **large sample sizes**.

## Analysis of Variance (ANOVA)

Used to test hypotheses on the **means** of three or more populations.

# The t-test

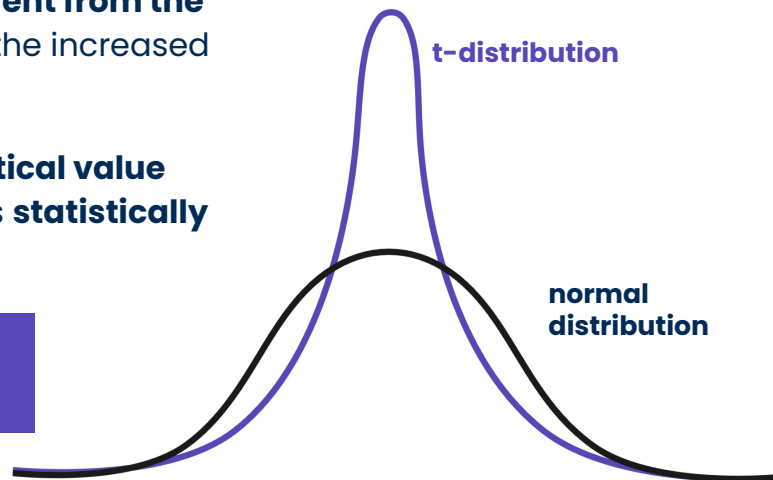
The t-test is a parametric test based on the **t-distribution** and is used to test hypotheses on the **mean of a single population** or the **difference between the means of two samples** when the sample size is smaller.

The **t-distribution**, or Student's t-distribution, is used to calculate the probability of obtaining a **sample mean** that is **different from the population mean**. It has heavier tails to account for the increased variability in **smaller sample sizes**.

The **t-test** uses the t-distribution to **calculate the critical value** that is used to determine if the **difference** between is **statistically significant**.

$|t\text{-score}| \geq \text{critical value} \rightarrow \text{Reject the null hypothesis}$

$|t\text{-score}| < \text{critical value} \rightarrow \text{Fail to reject the null hypothesis}$



# One-sample t-test

To compare a **sample mean** with the **population mean**, we use a **one-sample t-test**.

The **test statistic t (t-score)** is:

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

$\bar{x}$  is the sample mean  
 $s$  is the sample standard deviation  
 $n$  is the sample size  
 $\mu$  is the population mean

We can use the **AVERAGE()**, **STDEV()**, and **COUNT()** Google Sheet functions to calculate the t-score.

**Note:** We also need to **determine the critical value** using the degrees of freedom and level of significance either from a statistical table, online calculator, or using Google Sheets.

## Assumptions for a one-sample t-test:

- 01. Random sampling:** The data are collected using a random sampling method to ensure that the sample is representative of the population.
- 02. Normality:** The distribution of the sample means is approximately normal.
- 03. Independence:** The observations in the sample are independent of each other. In other words, the value of one observation is not related to the value of another observation.
- 04. Homogeneity of variance (homoscedasticity):** The variance of the sample is approximately equal to the variance of the population.

# Independent two-sample t-test

To compare the **means of two independent samples** when there is no link between the two groups, we use an **independent two-sample t-test**.

The **t-score** is:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s^2 \left( \frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$\bar{x}_1$  is sample one's mean  
 $\bar{x}_2$  is sample two's mean  
 $s$  is pooled standard deviation  
 $n_1$  is sample one's size  
 $n_2$  is sample two's size

The pooled standard deviation is:

$$s = \sqrt{\frac{(n_1 - 1) \times s_1^2 + (n_2 - 1) \times s_2^2}{n_1 + n_2 - 2}}$$

## Assumptions for independent samples t-test:

- 01. Normality:** The data in each group are normally distributed.
- 02. Homoscedasticity:** The variance of the data in each group is equal.
- 03. Independence:** The observations within each group are independent of each other, and the two groups are independent of each other.

**Note:** To find the p-value from the statistical table, we need to use the degrees of freedom as  $(n_1 + n_2 - 2)$  for the independent two-sample test.

# Paired two-sample t-test

To compare the **means of two related samples**, i.e., two sets from the same group, we use a **paired two-sample t-test**.

The **t-score** is:

$$t = \frac{\bar{d}}{\sqrt{\frac{s^2}{n}}}$$

$\bar{d}$  is the mean of the differences between the samples

$S$  is the standard deviation of the differences

$n$  is the sample size

## Assumptions for paired samples t-test:

- 01. Normality:** The differences between the paired observations are normally distributed.
- 02. Independence:** The paired observations are independent of each other.

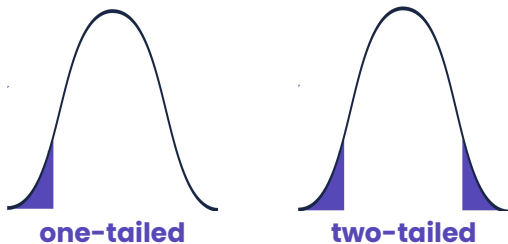
**Note:** To find the p-value from the statistical table, we need to use the degrees of freedom as  $(n - 1)$  for the paired two-sample test.

# T-tests in Google Sheets

We can either use `AVERAGE()`, `STDEV()`, and `COUNT()` to calculate the t-score using Google Sheets and a p-value table to determine the p-value, or we can use the built-in function `T.TEST()` to calculate the p-value.

`=T.TEST(range1, range2, tails, type)`

- **range1**—The first sample of data or group of cells to consider for the t-test.
- **range2**—The second sample of data or group of cells to consider for the t-test.
- **tails**—Specify the number of distribution tails.
  - If **1**: uses a one-tailed distribution.
  - If **2**, **uses** a two-tailed distribution.
- **type**—Specifies the type of t-test.
  - If **1**, a paired test is performed.
  - If **2**: a two-sample equal variance (homoscedastic) test is performed.
  - If **3**: a two-sample unequal variance (heteroscedastic) test is performed.



If the populations being compared have equal variance, then we can use the two-sample equal variance test. If the variances differ, we use the two-sample unequal variance test.

**Note:** The `T.TEST()` function output is the probability associated with the t-test, i.e., the **p-value**.



# The z-test

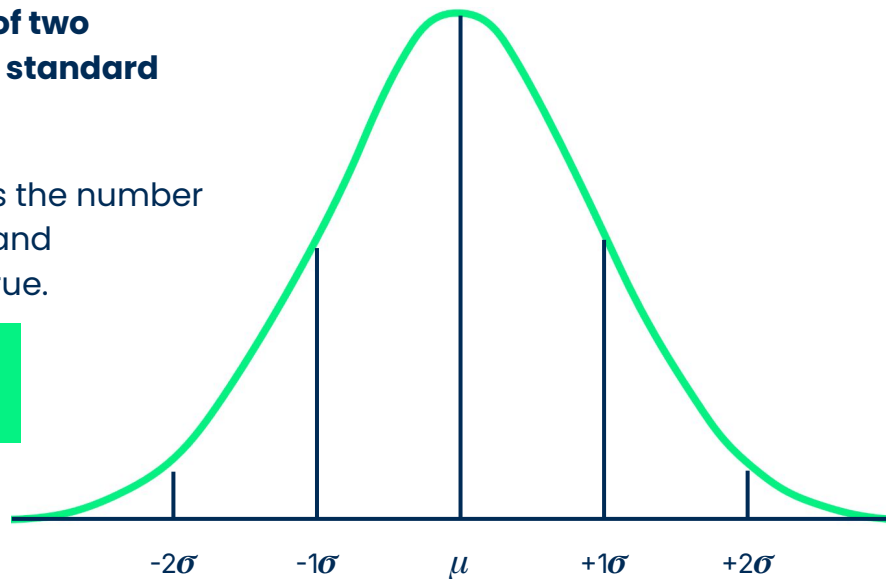
The z-test is a parametric test based on the normal distribution (a.k.a. the z-distribution) and is similar to the t-test. However, the z-test is used when the **sample is large** and the **population standard deviation is known**.

In the **z-test**, the difference between the **means of two populations** is expressed in terms of the **number of standard deviations** ( $\sigma$ ).

The **z-score** (test statistic  $z$ ) therefore represents the number of standard deviations between the sample mean and population mean, assuming the null hypothesis is true.

**$|z\text{-score}| \geq \text{critical value} \rightarrow \text{Reject the null hypothesis}$**

**$|z\text{-score}| < \text{critical value} \rightarrow \text{Fail to reject the null hypothesis}$**



# One-sample z-test

To compare a **sample mean** with the **population mean** when the sample size is large and the standard deviation is known, we use a **one-sample z-test**.

The **test statistic z** is:

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$\bar{x}$  is the sample mean  
 $\sigma$  is the **population standard deviation**  
 $n$  is the sample size  
 $\mu$  is the population mean

**Note:** The only difference between the test statistic  $t$  and  $z$  for a one-sample test is using the **sample** standard deviation for the  $t$  statistic and the **population** standard deviation for the  $z$  statistic.

## Assumptions for one-sample z-test:

01. **Random sampling:** The sample is selected randomly from the population.
02. **Normal distribution:** The population from which the sample is drawn is normally distributed.
03. **Large sample size:** The sample size is sufficiently large, typically at least 30, so that the central limit theorem can be applied.
04. **Independence:** The observations in the sample are independent of each other.
05. **Known population standard deviation:** The standard deviation of the population is known.

# Two-sample z-test

To compare the **means of two different samples** when the sample size is large and the standard deviation is known, we use a **two-sample z-test**.

The **test statistic z** for independent groups:

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$\bar{x}_1$  is sample one's mean  
 $\bar{x}_2$  is sample two's mean  
 $\sigma_1$  is population one's standard deviation  
 $\sigma_2$  is population two's standard deviation  
 $n_1$  is sample one's size  
 $n_2$  is sample two's size

The **test statistic z** for related groups:

$$z = \frac{\bar{d} - D}{\sqrt{\frac{\sigma^2}{n}}}$$

$\bar{d}$  is the mean of the differences between the samples  
 $D$  is the hypothesised mean of the differences (usually equal to zero)  
 $\sigma$  is the standard deviation of the differences  
 $n$  is the sample size

## Assumptions for two-sample z-test:

01. **Normality:** The data in each group are normally distributed.
02. **Homoscedasticity:** The variance of the data in each group is equal.
03. **Independence:** The two samples are independent of each other.
04. **Known population standard deviations:** The standard deviations of the populations are known.

# Z-tests in Google Sheets

We can either use `AVERAGE()`, `STDEV()`, and `COUNT()` to calculate the z-score using Google Sheets and a statistical table to determine the p-value, or we can use the built-in function `Z.TEST()` to calculate the p-value.

```
=Z.TEST(data, value, [standard_deviation])
```

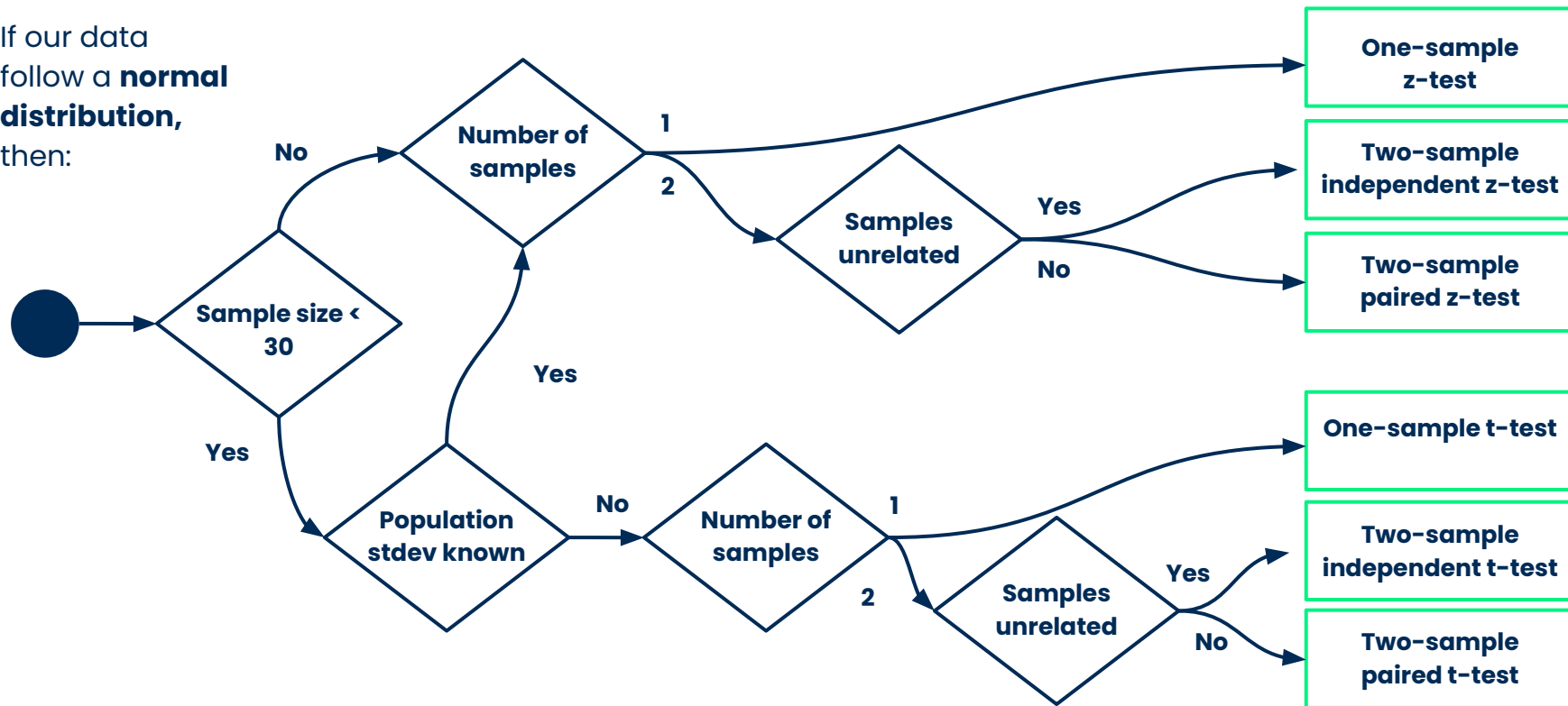
- **data**—The sample of data to consider for the z-test.
- **Value**—The test statistic to use in the z-test, most often the population mean.
- **[standard deviation]** – The standard deviation to assume for the z-test, often the standard deviation of the population.

**Note:** The `Z.TEST()` function output is the probability associated with the z-test, i.e., the **p-value**.

It is important to note how the **Google Sheets** hypothesis testing **functions differ** based on the test we need to use, whether it is a one- or two-sample test, and whether our samples are independent or paired.

# How to choose a parametric test

If our data follow a **normal distribution**, then:

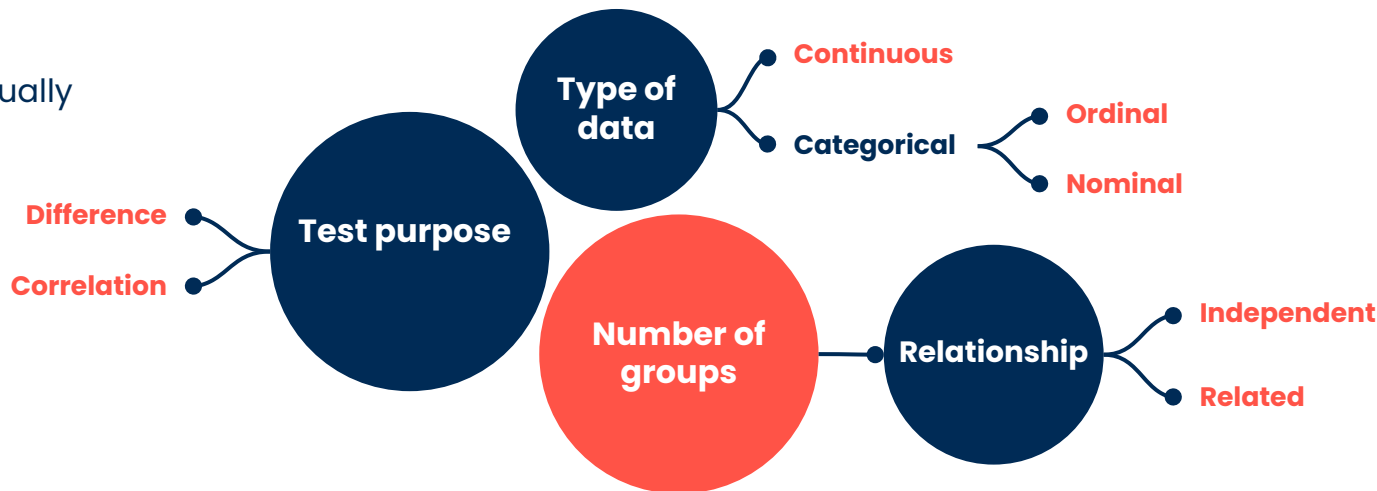


# Non-parametric tests overview

**Non-parametric tests** are a group of statistical tests that are used to **test hypotheses** when the **underlying distribution** of the data is **unknown**.

Several non-parametric tests are available to use in hypothesis testing.

**Which test to apply** usually depends on:



# Non-parametric tests overview

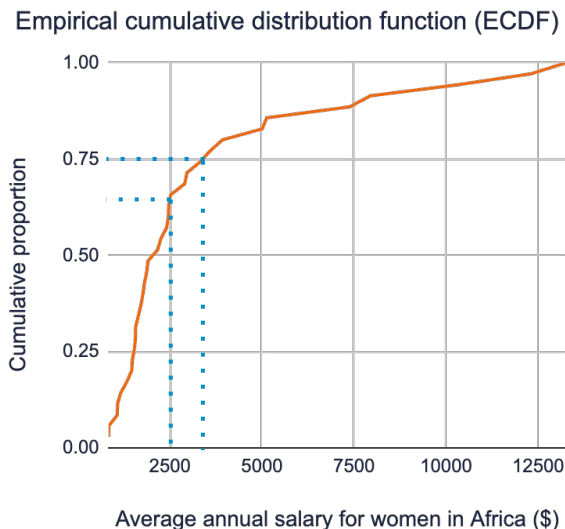
Rather than relying on estimates of population parameters as parametric tests do, non-parametric tests are **based on ranks** or the **number of times certain events occur** in the data.

Some of the most commonly used non-parametric tests include:

- **Kolmogorov-Smirnov test:** Compares the distribution of a sample with a theoretical distribution.
- **Mann-Whitney U-test:** Determines if two independent groups come from populations with the same distribution.
- **Chi-square:** Tests for independence between categorical variables in a contingency table.
- **Spearman's rank correlation coefficient:** Measures the strength and direction of the relationship between two variables using ranks.
- **Wilcoxon signed rank test:** Determines if there is a significant difference between two paired samples using ranks.
- **Friedman test:** Tests for significant differences between three or more paired samples using ranks.
- **Kruskal-Wallis H test:** Tests for significant differences between three or more independent groups using ranks.

# Kolmogorov-Smirnov and ECDF

Kolmogorov-Smirnov (KS) is a **non-parametric** test based on the **empirical cumulative distribution function** (ECDF), which is a way to visually represent how data are **distributed**.



The ECDF **maps each observation** in a dataset **to the proportion of observations** that are less than or equal to it.

It is a step function that increases by  $1/n$  at each point, where  $n$  is the sample size.

For example, considering the ECDF for the average annual salary for women in Africa, we see that more than 50% of women earn \$2500 or less per year. We also see that 75% of women earn less than \$3750 per year.



# Kolmogorov-Smirnov test overview

Kolmogorov-Smirnov (KS) is used to test hypotheses on whether an **underlying distribution** observed in a sample is **similar to the hypothesized distribution** or whether **two distributions are similar**.

Considering that KS helps us examine underlying distributions, it is a **useful tool to test for normality**, which is a prerequisite of parametric tests.

As with parametric tests, we need to state the **null** and **alternative hypotheses**:

- $H_0$  is that the sample is drawn from a population with a specific distribution, e.g., a normal distribution.
- $H_A$  is that the sample is not drawn from a population with the specified distribution.

We will also need to specify the **level of significance** ( $\alpha$ ), calculate a **test statistic** (denoted **D** for Kolmogorov-Smirnov), and determine the **critical value** and **p-value**.

**$|D| \geq \text{critical value} \rightarrow \text{Reject the null hypothesis}$**

**$|D| < \text{critical value} \rightarrow \text{Fail to reject the null hypothesis}$**

**$p\text{-value} \leq \alpha \rightarrow \text{Reject the null hypothesis}$**

**$p\text{-value} > \alpha \rightarrow \text{Fail to reject the null hypothesis}$**

# Kolmogorov-Smirnov test statistic

The **Kolmogorov-Smirnov test statistic** is:

$$D = \max_{1 \leq i \leq n} \left( \left| F(Y_i) - \frac{i-1}{n} \right|, \left| \frac{i}{n} - F(Y_i) \right| \right)$$

where

$i$  is the index of the ordered sample  $Y_1, Y_2$

is the sample size

$n$

the  $i$ th ordered value in the sample

$Y_i$

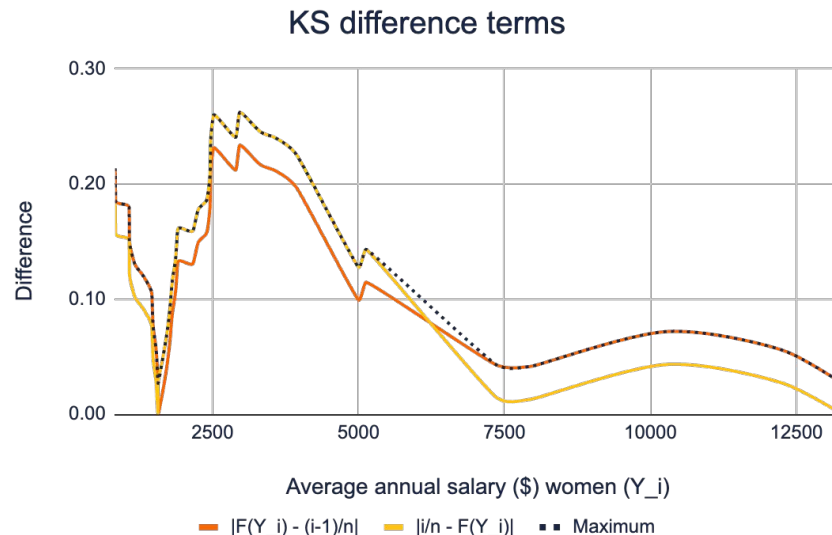
is the hypothesised cumulative distribution function (CDF)

$F(Y_i)$  evaluated at the  $i$ th ordered value of the sample data  $Y_i$

Both  $(i-1)/n$  and  $i/n$  represent the empirical cumulative distribution function (ECDF)\*.

The **test statistic D** is a single value which is the **maximum** across both difference terms,  $|F(Y_i) - (i-1)/n|$  and  $|i/n - F(Y_i)|$ , for all sample values,  $Y_i$ .

We need to calculate both difference terms to ensure that the ECDF starts at 0  $((i-1)/n)$  and ends at 1  $(i/n)$ , i.e., the entire range.



\* $(i-1)/n$  represents the cumulative proportion of observations that are expected to be strictly less than the  $i$ th ordered value, while  $i/n$  represents the proportion that is less than or equal to the  $i$ th.

# Steps to the Kolmogorov-Smirnov test

## The steps to performing KS:

01. State the **null** and **alternative** hypotheses:
  - a.  $H_0$  is that the sample is drawn from a population with a specific distribution, e.g., a normal distribution.
  - b.  $H_A$  is that the sample is not drawn from a population with the specified distribution.
02. Specify the **level of significance** ( $\alpha$ ).
03. Calculate the **test statistic**,  $D$ , using the Kolmogorov-Smirnov test statistic formula.
04. Determine the **critical value** using the KS table, level of significance, and sample size.
05. Compare the test statistic ( $D$ ) to the critical value.

Although the **p-value** for a KS test can be calculated using statistical software or a programming language like Python using built-in functions, it is much more involved and resource intensive in Google Sheets.

In theory, the p-value can be calculated using either the **exact** method, when  $n \leq 35$ , or the **approximate** (also known as asymptotic) method for larger sample sizes ( $n$ ).